

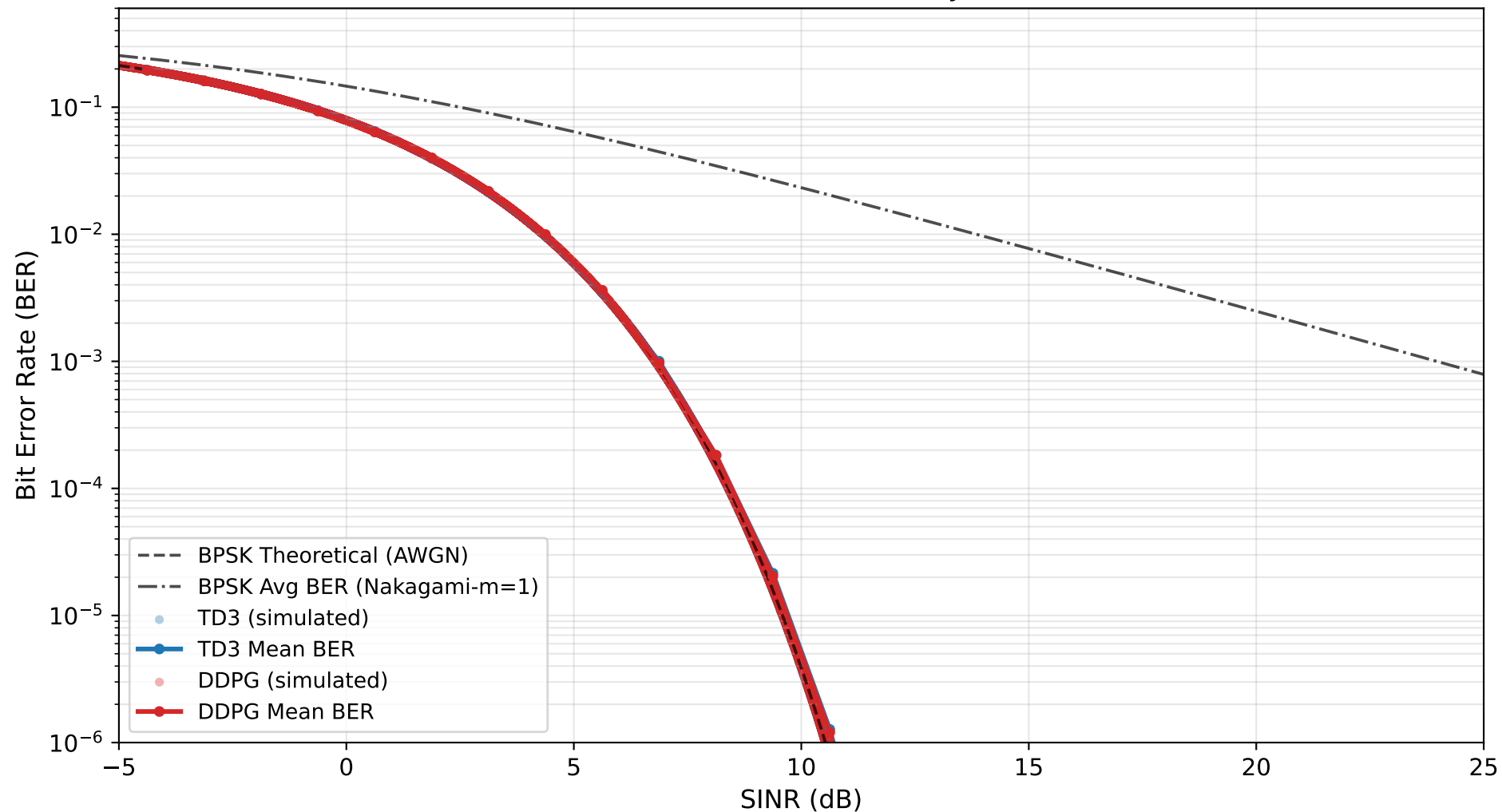
CRN Power Control: TD3 vs DDPG

Nakagami-m Fading Channel Performance Report

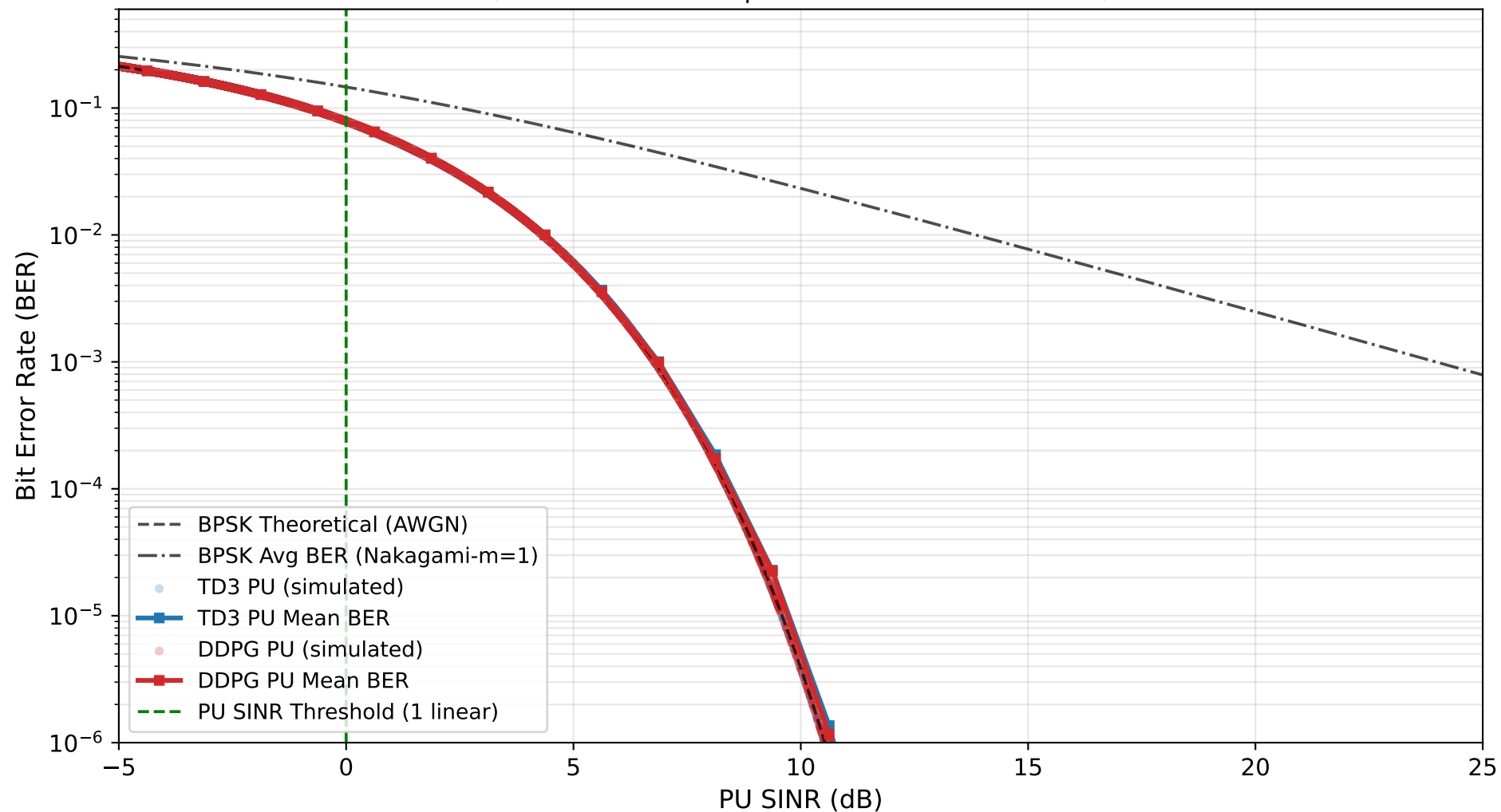
Nakagami-m = 1.0 | Episodes = 3500 | Steps/Episode = 200 | SINR Threshold = 1.0 dB

Metric	TD3	DDPG	Winner
Avg Reward (last 100 ep)	310.7469	272.0717	TD3
SU Throughput (bits/s/Hz)	2.3652	2.0650	TD3
PU Throughput (bits/s/Hz)	0.7853	0.9991	DDPG
Outage Probability	0.2491	0.3210	TD3
Average BER	0.058879	0.076984	TD3
Training Time (s)	24163.4	22597.9	-

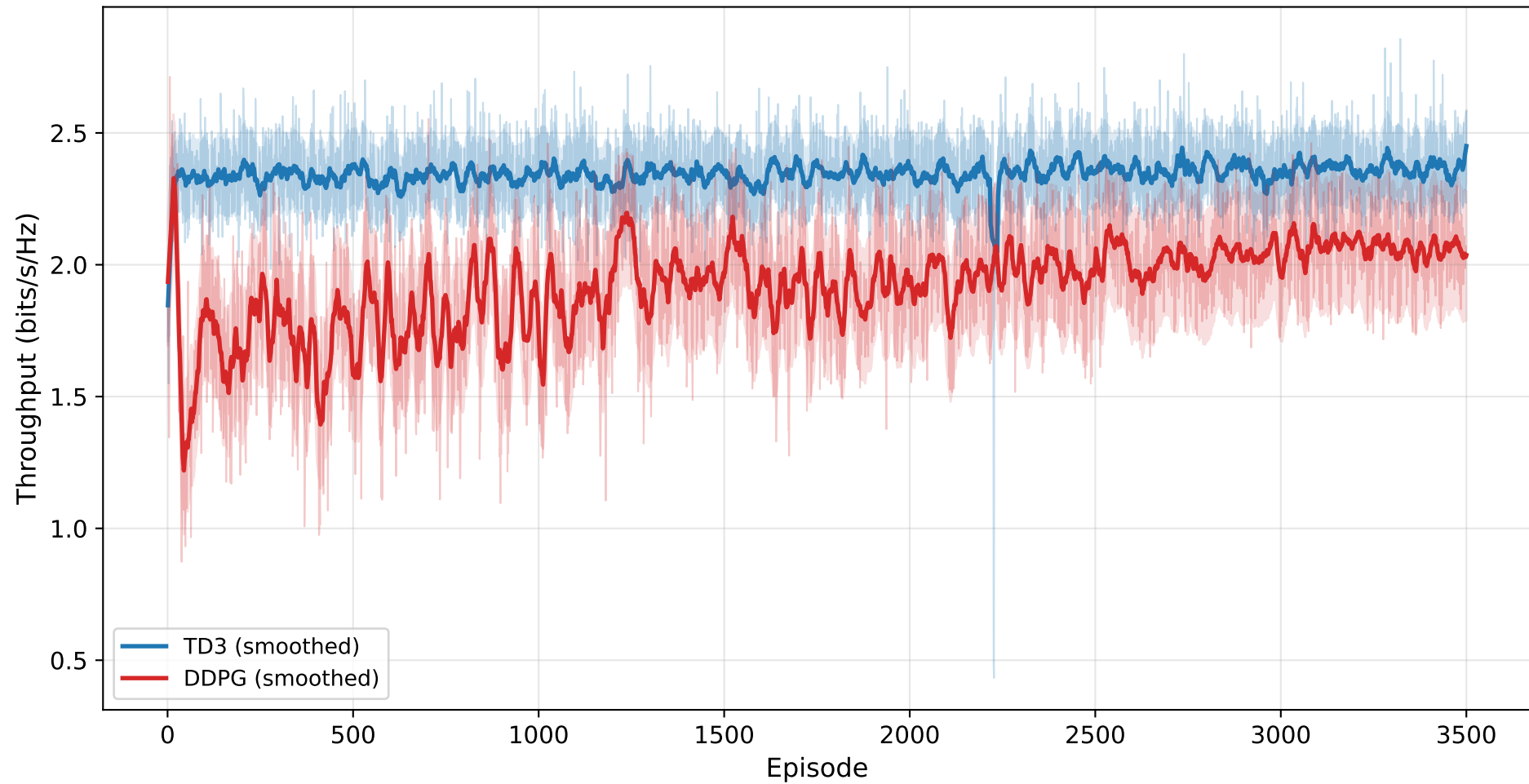
SINR vs BER - Nakagami-m=1 Fading Channel
(BPSK Modulation, Secondary User)



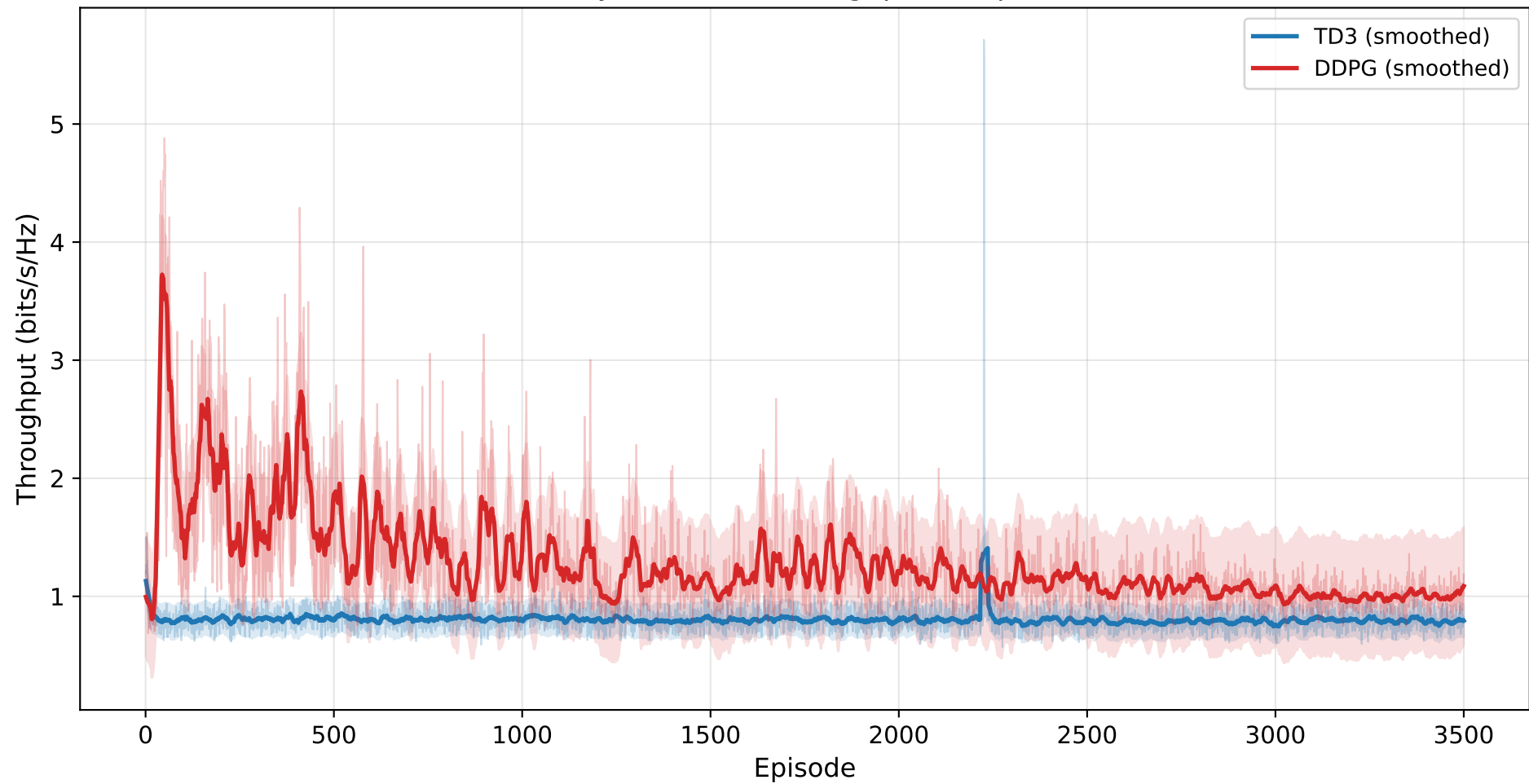
Primary User SINR vs BER - Nakagami-m=1 Fading Channel
(BPSK Modulation | Effect of SU Interference)



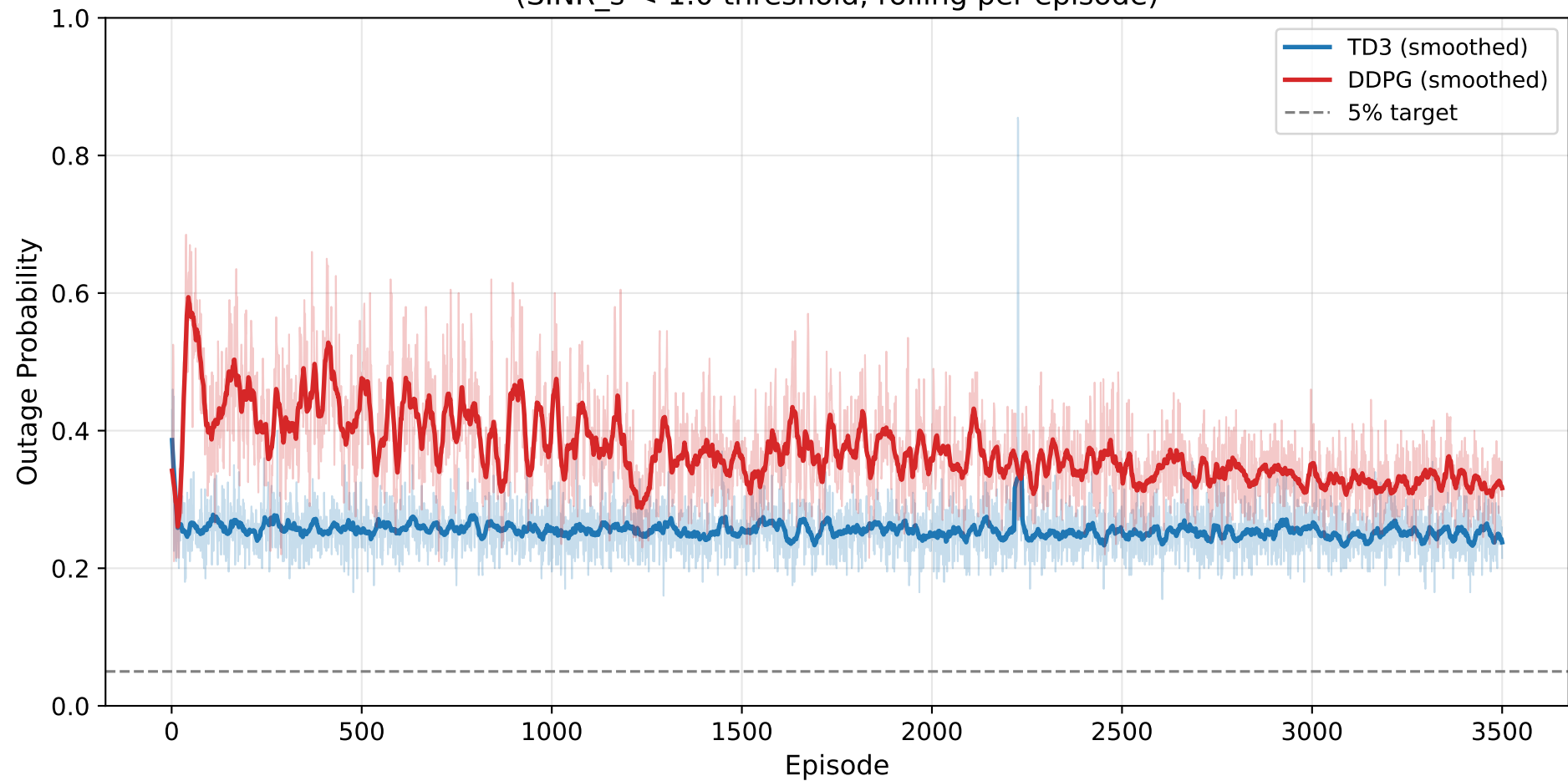
Secondary User (SU) Throughput vs Episodes



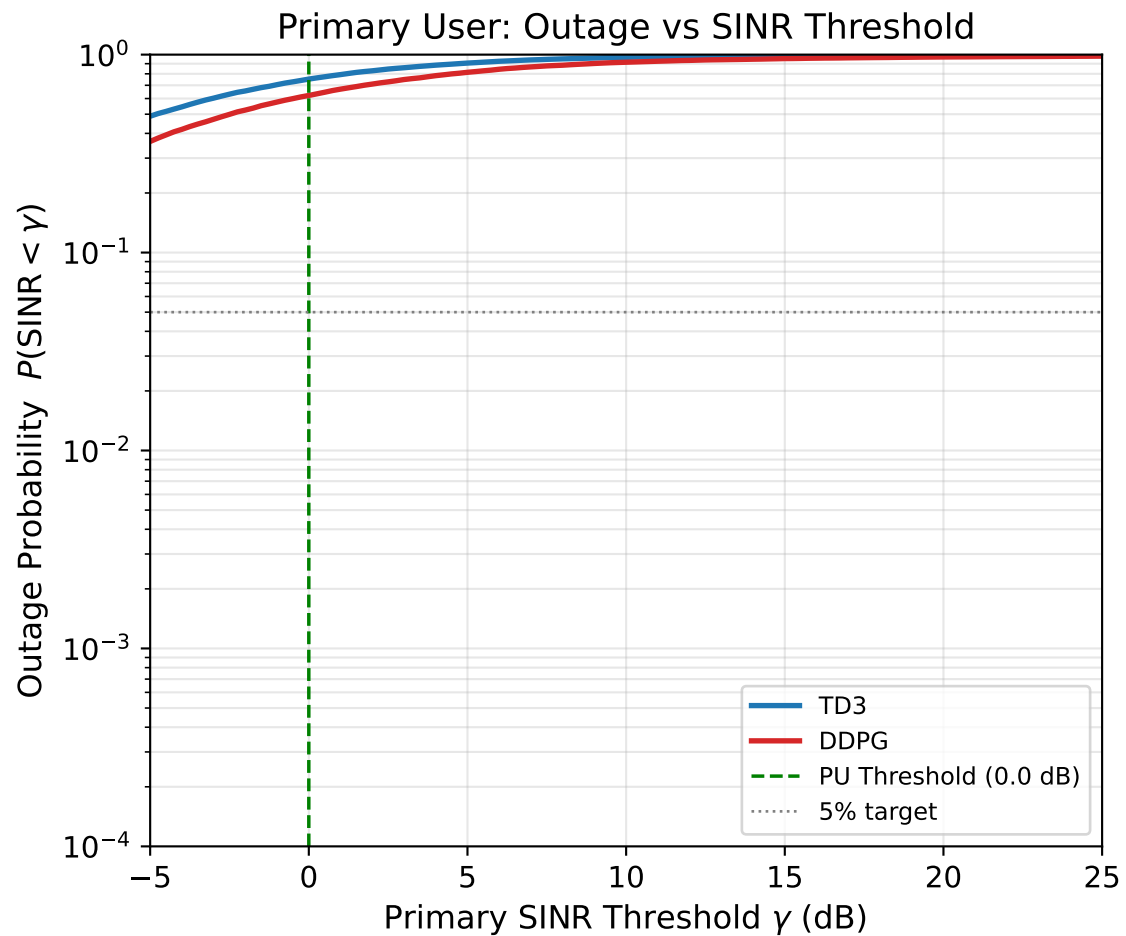
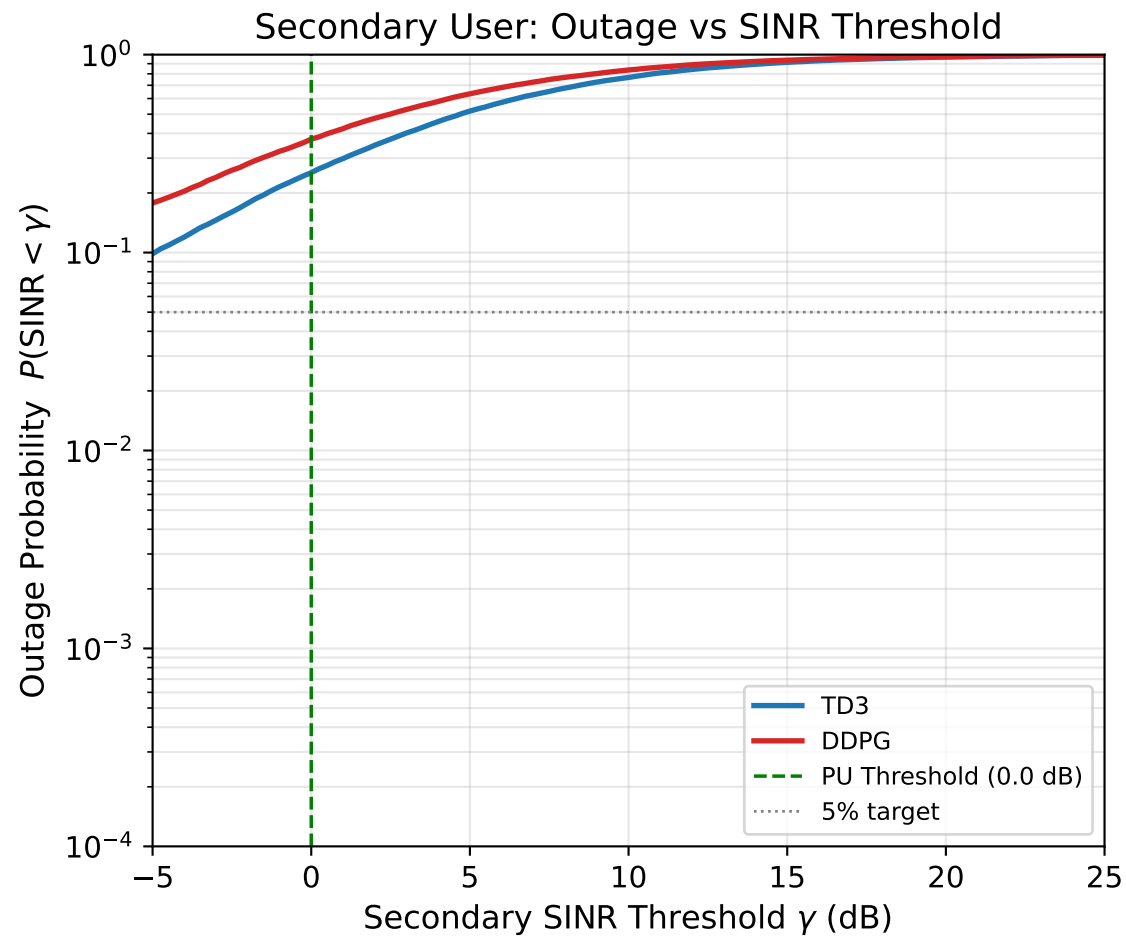
Primary User (PU) Throughput vs Episodes



Outage Probability vs Episodes
(SINR_s < 1.0 threshold, rolling per episode)

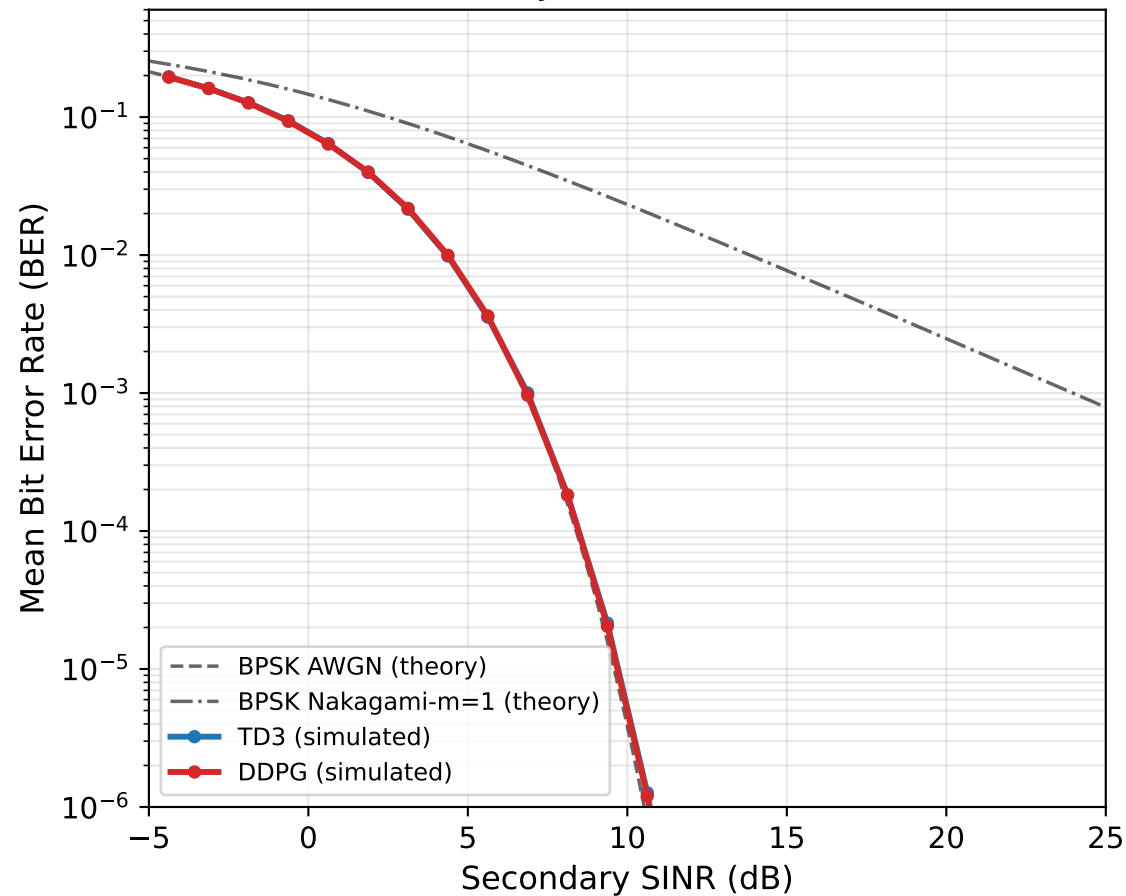


Outage Probability vs SINR Threshold (Empirical CDF) — Imperfect CSI

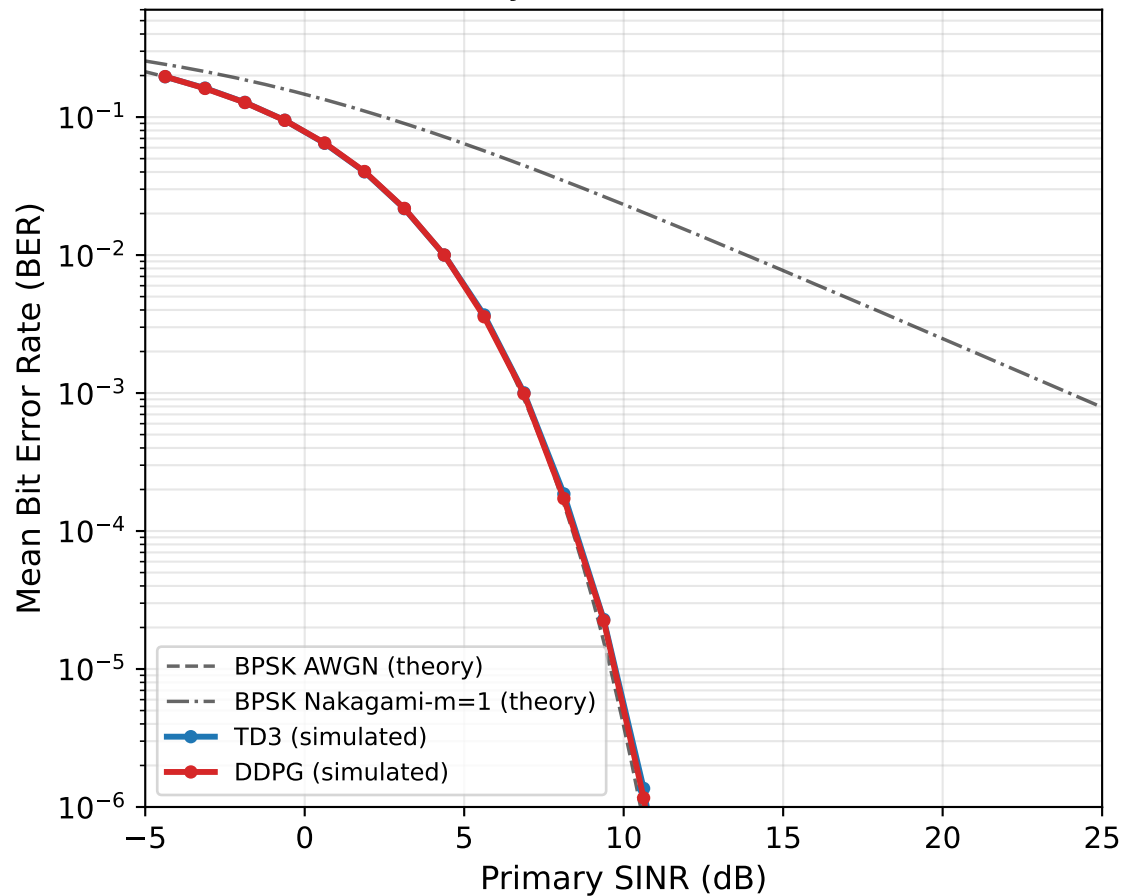


BER vs SINR — Algorithm Comparison under Imperfect CSI

Secondary User: BER vs SINR

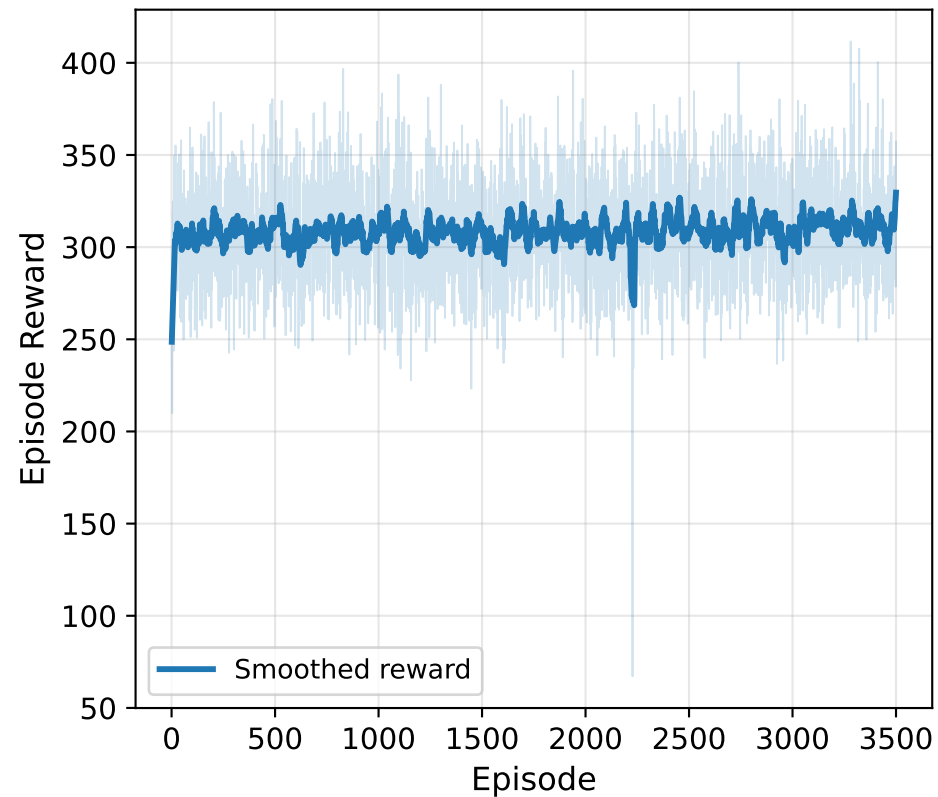


Primary User: BER vs SINR

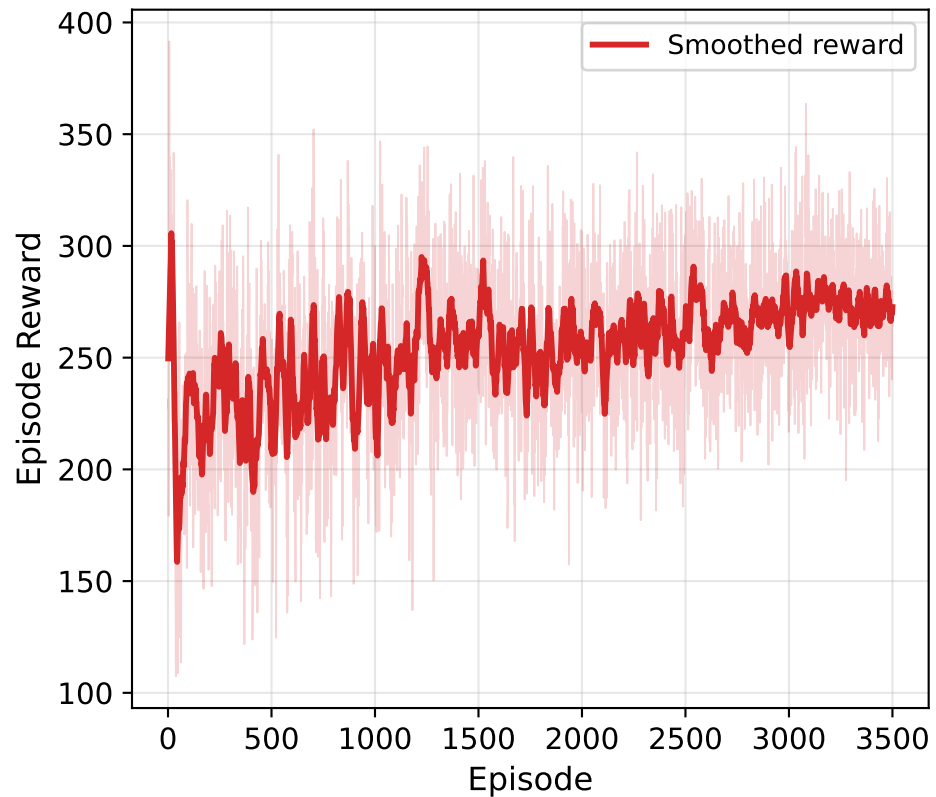


Episode Reward Curves - TD3 vs DDPG

TD3 Reward Curve



DDPG Reward Curve



TD3 vs DDPG - Reward Comparison

